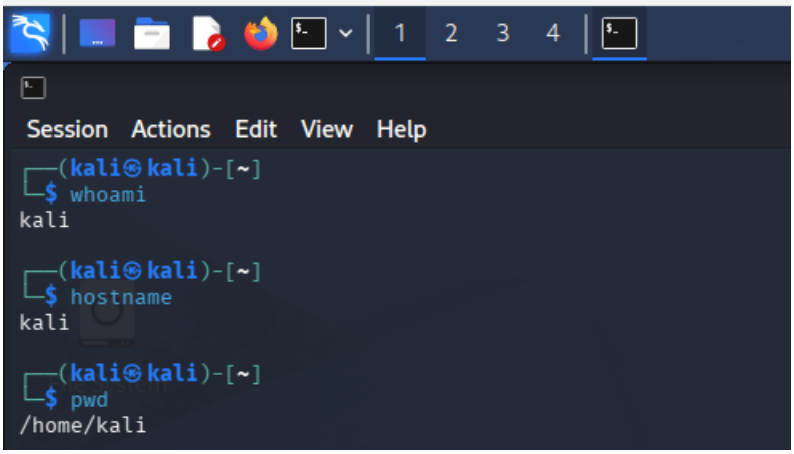
PART 1 – **SYSTEM INFORMATION**

Step 1: ***Who am I and where am I?***



→Username: kali

→Machine hostname: kali

→Directory: /home/kali

Step 2: ***Operating System & Hardware Info.***

```
(kali㉿kali)-[~]
$ uname -a
Linux kali 6.19.14+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.19.14-1+kali1 (2026-05-05) x86_64 GNU/Linux
```

```
(kali㉿kali)-[~]
$ free -h
               total        used        free      shared  buff/cache   available
Mem:           1.9Gi       795Mi       560Mi        14Mi       769Mi       1.1Gi
Swap:          953Mi          0B       953Mi

(kali㉿kali)-[~]
$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            889M     0  889M   0% /dev
tmpfs           197M  964K  196M   1% /run
/dev/sda1        79G   16G   59G  21% /
tmpfs           985M  4.0K  985M   1% /dev/shm
none            1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
tmpfs           985M   36K  985M   1% /tmp
none            1.0M     0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs           197M  108K  197M   1% /run/user/1000
```

→Total VM RAM: 1.9Gi

→Disk space available on main partition: 59G

→Version of Linux Kernel: 6.19.14+kali-amd64

PART 2 – **BASIC NETWORK CONNECTIVITY TEST**

Step 1: ***Ping a known IP address***

```
┌──(kali㉿kali)-[~]
└─$ ping -c2 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=64 time=7.87 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=64 time=8.01 ms

── 8.8.8.8 ping statistics ──
2 packets transmitted, 2 received, 0% packet loss, time 1001ms
rtt min/avg/max/mdev = 7.865/7.939/8.013/0.074 ms
```

→The average round-trip time: 7.939 ms

→There was 0% packet loss.

Step 2: ***Ping a Domain Name***

```
┌──(kali㉿kali)-[~]
└─$ ping -c4 google.com
PING google.com (142.250.31.113) 56(84) bytes of data.
64 bytes from bj-in-f113.1e100.net (142.250.31.113): icmp_seq=1 ttl=64 time=12.0 ms
64 bytes from bj-in-f113.1e100.net (142.250.31.113): icmp_seq=2 ttl=64 time=10.4 ms
64 bytes from bj-in-f113.1e100.net (142.250.31.113): icmp_seq=3 ttl=64 time=8.84 ms
64 bytes from bj-in-f113.1e100.net (142.250.31.113): icmp_seq=4 ttl=64 time=10.6 ms

── google.com ping statistics ──
4 packets transmitted, 4 received, 0% packet loss, time 3010ms
rtt min/avg/max/mdev = 8.844/10.461/11.961/1.104 ms
```

→IP address of google.com ---> 142.250.31.113

→The round-trip time is different from pinging 8.8.8.8

→Round-trip times differ because network conditions, routes, and device processing loads change with each packet.

→-c 4 means the number of packets transmitted was 4. That means 4 ping requests were sent.